Course Code	EE 201	**Course Title**	Circuit Theory I
Type of Course	☒ Compulsory ☐ Elective	**Semester**	☐ Fall ☒ Spring ☐ Summer
Credit Hours / ECTS credits	(3+0+2) 4 Credits / 8 ECTS	**Pre-requisite**	MATH 101 AND PHYS 101 (or PHYS 105)
		Co-requisite	NONE
Mode of Delivery	☒ Face-to-face ☐ Distance learning	**Language of Instruction**	☒ English ☐ Turkish
Course Level	☒ Undergraduate ☐ Graduate (MS) ☐ Graduate (PhD)	**Year of Study**	☐ First Year ☒ Sophomore ☐ Junior ☐ Senior ☐ Masters ☐ Doctorate

Course Coordinator	Prof. Dr. A. Nezih Güven E-mail: nezih.guven@tedu.edu.tr | Room: F-318	
Office Hours	By appointment via e-mail	
Teaching Assistant(s)	Gülce Turhan, E-mail: gulce.turhan@tedu.edu.tr, Room: A324	
Office Hours	To be determined	
Course Schedule	Section 01	Monday (11:00–12:50) @F413 Wednesday (13:00–13:50) @F413 Lab: Group 1: Thursday (16:00–17:50) @A317-L Group 2: Friday (10:00–11:50) @A317-L

Catalog Description	Fundamentals of electric circuits, variables, and lumped circuit elements. Kirchhoff's laws. Resistive circuits. Methods of circuit analysis. Operational amplifiers (op amps). Energy storage elements. Analysis of first and second-order circuits. Experiments on resistive circuits, first and second-order circuits, and op amp circuits.
Course Objectives	This course equips students with the fundamentals of electric circuits, including energy storage, transient and steady-state responses, and operational amplifiers. The course covers the fundamental laws and techniques, such as nodal and mesh analysis and Thevenin and Norton equivalent circuits.
Required Reading	Textbook: Nilsson, J.W. & Riedel, S. A. (2020). Electric Circuits. 11 th Ed., Pearson. *Note: 9th and 10th editions of this book are also acceptable.*
Suggested / Recommended Reading	1. Johnson, D. E., Johnson, J. R., Hilburn, J. L. & Scott, P. D. (1997). Electric Circuit Analysis. 3rd Ed., John Wiley & Sons. 2. Alexander, C. K., & Sadiku, M. (2009). Fundamentals of Electric Circuits. 4th Ed., McGraw Hill. 3. Hayt, K., Kemmerly, J. & Durbin, S. (2018). Engineering Circuit Analysis. 9th Ed., McGraw-Hill Education.
Software Usage	MATLAB and LTspice (https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html)
Course Learning Outcomes (LOs)	Upon successful completion of the course, students will be able to: 1. Recognize the basic circuit concepts, including voltage, current, energy, and power, 2. Express the node and mesh analysis methods used in linear time-invariant circuits analysis,

	3. Apply Thevenin and Norton theorems for investigating circuits, including operational amplifiers, 4. Analyze alternating current (AC) circuits using sinusoidal steady-state analysis techniques, 5. Evaluate the behavior of capacitors and inductors in circuits, assessing both transient and steady-state responses of first-order circuits, 6. Experiment on resistive circuits, first and second-order circuits, and op amp circuits in collaboration with peers.
Course Evaluation	Course feedback survey will be conducted in the last two weeks of the semester.
Course Web Page	The class has already been enrolled in Moodle (https://lms.tedu.edu.tr/). All announcements and course-related materials will be posted on the Moodle course page.
Some Useful Website References	N/A

Learning Activities & Teaching Methods	☒ Brainstorming ☐ Case Study/Scenario Analysis ☒ Collaborating ☐ Concept Mapping ☒ Demonstrating ☒ Discussions / Debates ☐ Drama / Role Playing ☒ Experiments ☐ Field Trips ☐ Guest Speakers	☒ Hands-on Activities ☒ Inquiry ☐ Microteaching ☐ Oral Presentations / Reports ☐ Peer Teaching ☐ Predict-Observe-Explain ☒ Problem Solving ☒ Questioning ☒ Reading	☐ Scaffolding / Coaching ☐ Seminars ☐ Service Learning ☐ Simulations & Games ☒ Telling / Explaining ☐ Think-Pair-Share ☐ Video Presentations ☒ Web Searching ☐ Other(s): ……………

Assessment Methods & Criteria	☒ Case Studies / Homework	**(9%)**	☐ Presentation (Oral, Poster)	(...%)
	☒ Lab Assignment	**(20%)**	☐ Project	(...%)
	☐ Observation	(...%)	☒ Quiz	**(4%)**
	☐ Oral Questioning	(...%)	☐ Self-evaluation	(...%)
	☐ Peer Evaluation	(...%)	☒ Test/Exams	**(67%)**
	☐ Performance Project (Written, Oral)	(...%)	☐ Other(s): ……………	(...%)
	☐ Portfolio	(...%)		

Student Workload	☐ Case Study Analysis	(... hrs)	☐ Online Discussion	(... hrs)
	☒ Course Readings	**(50 hrs)**	☐ Oral Presentation	(... hrs)
	☐ Debate	(... hrs)	☐ Poster Presentation	(... hrs)
	☐ Demonstration	(... hrs)	☒ Report on a Topic	**(16 hrs)**
	☒ Exams/Quizzes	**(50 hrs)**	☐ Research Review	(... hrs)
	☐ Field Trips/Visits	(... hrs)	☐ Resource Review	(... hrs)
	☐ Hands-on Work	(… hrs)	☐ Team Meetings	(... hrs)
	☒ Lab Applications	**(36 hrs)**	☐ Web Designs	(... hrs)

	☒ Lectures	**(42 hrs)**	☐ Work Placement	(... hrs)
	☐ Mock Designs	(... hrs)	☐ Workshop	(... hrs)
	☐ Observation	(... hrs)	☒ Other(s): Problem solving	**(46 hrs)**
			Total Workload	**240 hrs**

TENTATIVE OUTLINE

Any changes and updates will be announced on the course web page.

Week	Topics	LOs	Textbook Reading	Assignments
1	Introduction to Electrical Circuits	1	Ch. 1	
2	Resistive Circuits; Sources; Measurement Equipment ***Lab: Introduction to Circuits Lab***	1	Ch. 2, 3	
3	Power; Nodal Analysis	1, 2	Ch. 2, 3, 4	HW #1
4	Nodal Analysis ***Lab#1 Introduction to Resistive Circuits***	2	Ch. 4	Read Appendix A
5	Mesh Analysis ***Lab#2 Introduction to Mixed Signal Oscilloscope***	2	Ch. 4	HW#2
6	Thevenin's and Norton's Theorems; Superposition; Source Transformation; Maximum Power Transfer	3	Ch. 4	
7	Operational Amplifiers ***Lab#3 DC and AC Analysis of Resistive Circuits***	3, 4	Ch. 5	**MT #1** HW#3
8	Analysis of Resistive Op-Amp Circuits	3, 4	Ch. 5	
9	Energy-Storage Elements ***Lab#4 Operational Amplifiers***	5	Ch. 6	
10	First-Order Circuits ***Lab: Seminar (Occupational Health and Safety)***	5	Ch. 7	HW #4
11	First-Order Circuits	5	Ch. 7	
12	Second-Order Circuits	5	Ch. 8	
13	Second-Order Circuits ***Lab#5 First-Order Circuits***	5	Ch. 8	**MT #2**

14	Second-Order Circuits ***Lab#6 Second-Order Circuits***	5	Ch. 8	HW #5
15	Introduction to Sinusoidal Steady-State Analysis	4	Ch. 9	
FINAL EXAMS WEEK (*date and time to be announced later*).				

COURSE ASSIGNMENTS & GRADING

A. Midterm Exams [40%]

There will be two closed-book & closed-note midterm exams.

B. Final Exam [27%]

There will be a closed-book final exam covering all topics. The date, place and time of the final exam will be announced at the end of the semester.

C. Quizes (4%)

D. Lab Experiments (20%): There will be a Final Lab Evaluation during the final exam period.

E. Homework Assignments [9%]:

There will be **at least** 4 homework assignments.

COURSE ASSESSMENTS & LEARNING OUTCOMES MATRIX

Assessment Methods	Course Learning Outcomes
Quizzes	LO# 1–6
Homeworks	LO# 1–6
Midterm Exams	LO# 1–6
Final Exam	LO# 1–6
Lab Work	LO# 1–6

COURSE POLICIES

I. Attendance

- Regular class attendance is expected for all students at the University. You are not required but strongly advised to attend all classes. The instructor may not accept you to the Final Exam if your overall class attendance is less than 50% and you will be given a failing grade in that case.
- Attendance to all lab sessions is mandatory.
- Please send your instructor a brief e-mail to explain your absence in advance.
- Your absence will not reduce your attendance rate *if and only if* you have a legitimate reason for missing a class (such as illness, death in the family, a traffic accident, *etc.*). In case of a disease or emergency, you must supply formal documentation that supports your claim.
- Classes start at the hour. Please be respectful of your classmates by being on time.
- All electronic equipment should be turned off and kept out of sight before the lecture starts.

II. Make-up Policy

There will be one make-up exam for the exams which will be available if and only if you have a legitimate reason for missing the exam (such as illness, death in the family, a traffic accident, *etc.*). In case of an illness or emergency, you must supply a formal documentation that supports your claim. The topics for the make-up exam are from everything that is covered in class at the time of the exam. There will be *no* make-up for the quizzes or homeworks.

III. Late Submission Policy

Late submissions will not be graded. Missed quizzes without any clarification will result in a grade of zero.

IV. Cheating & Plagiarism

Collaboration is strongly encouraged; however, the work you hand in must be solely your own. Cheating and plagiarism are severe offenses and will be penalized accordingly by the university disciplinary committee. Cheating has a comprehensive description which can be summarized as "acting dishonestly." Some of the things that can be considered cheating are the following:

- Copying answers on exams, home works and lab works,
- Using prohibited material on exams,
- Lying to gain any advantage in class,
- Providing false, modified, or forged data in a report,
- Plagiarizing (see below),
- Modifying graded material to be re-graded,
- Causing harm to colleagues by distributing false information about an exam, homework, or lab.

All of the following are considered plagiarism (www.plagiarism.org):

- Turning in someone else's work as your own,
- Copying words or ideas from someone else without giving credit,
- Failing to put a quotation in quotation marks,
- Giving incorrect information about the source of a quotation,
- Changing words but copying the sentence structure of a source without giving credit,
- Copying so many words or ideas from a source that it makes up the majority of your work, whether you give credit or not.

V. Disability Support

If you have a disabling condition that may interfere with your ability to complete this course successfully, please contact Dr. A. Nezih Güven (nezih.guven@tedu.edu.tr).

VI. Student Development and Psychological Counseling Center

Student Development and Psychological Counseling Center conducts individual psychological counseling, group psychological counseling, preventive and developmental services such as workshops and seminars for all students in need. You may apply to our Center in order to deal with all your current problems.
For General Information: https://csc.tedu.edu.tr/
For Application: https://anket.tedu.edu.tr/index.php/761882?lang=en

VII. TEDU Without Barriers Unit

Please inform the TEDU Without Barriers Unit and the instructor of the course about the specific issues in case you have a physical or mental disability and are having trouble with anything related to this course such as accessing the material, participating in the class, taking notes, preparing for, attending or managing to complete the exams. Your situation will be reviewed by commission, in accordance with the principle of confidentiality, and if deemed appropriate, facilitating measures will be taken so that you can take the course more efficiently. For further information and/or questions:
engelsiz@tedu.edu.tr and https://www.tedu.edu.tr/engelsiz-tedu

Prepared By	Prof. Dr. A. Nezih Güven	**Date**	09/02/2026
Revised By		**Rev. Date**	